Factoring & Solving Polynomial Equations

Review: Factor: A) $t^2 - 10t + 25$ B) $4x^2 - 9$

**We will now factor polynomials with degrees higher than 2.

Sum of Two Cubes: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

Example 1: Factor:

A) $x^3 + 27$ B) $125 + 64x^3$

Difference of Two Cubes: $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Example 2: Factor:

A) $8x^3 - 27$ B) $8a^6 - 125b^3$

Quadratic Form

Example 3: Factor:

A) $x^2 - 36$ B) $25x^4 - 36$

C) $g^4 - 10g^2 + 25$ D) $a^2b^2 - 8ab^3 + 16b^4$

Factoring by Grouping:

Example 4: Factor:

A) $ra + rb + sa + sb$

B) $x^2y^2 - 3x^2 - 4y^2 + 12$

C) $bx^2 + 2a + 2b + ax^2$

Zero Product Property: If $AB = 0$, then $A = 0$ or $B = 0$

**can be used for more than two factors: i.e. If $ABCD = 0$, then $A = 0$ or $B = 0$ or $C = 0$ or D = 0

Example 5: Solve:

A) $2y^5 - 18y = 0$

B) $g^4 - 10g^2 = -25$

Example 6: You are building a bin to hold cedar mulch for your garden. The bin will hold 162 ft³ of mulch. The dimensions of the bin are x ft. by 5x – 6 ft. by 5x – 9 ft. Find the dimensions of the bin. (Draw a picture if it helps)